ABSTRACT

Online and hybrid education has made the webcam the main link between student and teacher, yet a teacher facing dozens of video tiles cannot tell when a learner loses focus. Student's engagement recognition task aims to solve this problem by classifying a short webcam clip into one of four ordinal levels, from Highly-Disengage to Highly-Engage. The task is hard because the labels are ordinal and severely imbalanced toward the higher engagement levels. Moreover, the clips are recorded under camera angles that differ from one student to the next. Prior work follows two approaches, using facial-behaviour toolkits like OpenFace or training from scratch with deep convolutional models like EfficientNetV2. Datasets for this task are limited and highly imbalanced like CMOSE and DAiSEE. I identify two limits of current approaches: the use of a single encoder for all facial features and the reliance on accuracy as the primary evaluation metric. In this thesis I keep the structure of the facial signal instead of flattening it, because gaze, landmarks, head pose, and action units move on different timescales. Additionally, I judge every model by quadratic-weighted kappa (QWK) rather than accuracy, because only an ordinal, imbalance-aware metric reflects real deployment conditions. My solution is a feature-group hybrid that decomposes the 709 OpenFace features into five behavioural groups, matches each group with its own temporal encoder (Temporal Convolutional Network, LSTM, or Transformer), and optionally adds a global I3D motion stream. Each stream has a small auxiliary head and their embeddings are recomposed to make the final prediction. I make three contributions. First, I ran an ablation across all 243 encoder assignments. It shows that the best configuration reaches a QWK of 0.605 on CMOSE against 0.537 for the best single-encoder baseline. Second, I collected and manually labelled a private webcam test set built with the same pipeline and reserved for testing. Third, I conducted a 3×3 cross-dataset study. It shows that models transfer poorly (off-diagonal QWK collapsing toward zero while accuracy remains acceptable) and the hybrid model trained on both datasets gives the best result on the unseen private set (QWK 0.379 against 0.285 for the best baseline).

Keywords: student engagement recognition, ordinal classification, class imbalance, quadratic-weighted kappa, facial-behaviour features, temporal modelling, cross-dataset generalisation.

Student

(Signature and full name)